Case 48: WU-E uniform-community spatial convergence

1 Verification purpose

This case verifies spatial convergence of the Wildland–Urban ELMFIRE (WU–E) ellipse–heat propagation model in a uniform idealized community. It holds the physical domain, initial burning area, wind, design-fire curve, linear characteristic building dimension, and separation fixed while refining the raster spacing through 20, 10, 5, and 2.5 m. The question is whether physically located ignition times, the community propagation rate, and domain-integrated received heat approach grid-independent values.

Current result: NOT EVALUABLE. 0 of 4 required simulation conditions have evaluable evidence. Analysis procedure: **INCOMPLETE.**

2 Mathematical formulation and numerical implementation

The uniform-community context follows the published WU–E parametric study [1]. Executable semantics were checked against ELMFIRE spread-rate calculation and ELMFIRE level-set formulation at source revision `a2dfbcdf7220`. The reviewed components were identical to that revision; this identity claim does not extend to other modified model components. The intended downscaled model interprets characteristic building dimension as the linear characteristic building dimension L_c in metres. For each spacing it uses

$$\beta_{ssd} = \min\left(1, \frac{s_{sd} + L_c}{\Delta x}\right), \qquad g_f = \min\left[\left(\frac{\Delta x}{L_c}\right)^2, 1\right], \qquad R_{search} = \left\lceil \frac{HAZ}{\Delta x} \right\rceil. \tag{1}$$

For a target cell receiving direct-flame-contact (DFC) and radiative fluxes, the cell-step energy is

$$E = (q_{\text{DFC}} + q_{\text{rad}})\, \Delta t\, \Delta x^2. \tag{2}$$

The material fire-thermal property (FTP) is fixed, with nominal $F_{TP,crit} = 5400\ \text{kJ m}^{-2}$, and the absolute urban spread magnitude is

$$U = \frac{60E}{0.3048\, \Delta t\, \Delta x\, F_{TP,crit}}. \tag{3}$$

WU-E then partitions this magnitude among head, flank, and backing directions using the wind- and geometry-dependent ellipse before ELMFIRE advances the level set with its two-stage integration. The pinned source omits β_{ssd} and g_f, uses a cell-count heat band, and replaces FTP using a 30,000-kJ branch; these are expected convergence defects, not reference calculation terms.

3 Derivation of expected behavior

Let Q_h denote any physical observable calculated on spacing h. A convergent discretization has

$$Q_h = Q + Ch^p + o(h^p), \qquad p \approx \frac{\log\left(\|Q_{2h} - Q_h\| / \|Q_h - Q_{h/2}\|\right)}{\log 2}. \tag{4}$$

Arrival times are evaluated at fixed physical landmarks by interpolation of the cross-community median time of arrival (TOA) profile; community rate of spread (ROS) is $1/b$ from $t = a + bx$ over ignited

downwind columns. Total received heat is the sum of ELMFIRE's terminal DFC and radiation energy rasters, whose stored values already include $\Delta t\,\Delta x^2$.

The four prepared grids use $h = 20, 10, 5, 2.5$ m with one fixed $\Delta t = 0.25$ s, so temporal-model and integration effects are not deliberately co-refined with space. Feedback-level timestamps must also show the complete 1,200-step sequence at that configured step. For a scalar Q, the finest-pair change is

$$e_Q = \frac{|Q_5 - Q_{2.5}|}{\max(|Q_{2.5}|, 10^{-12})};$$

for the three-component landmark vector, absolute values are replaced by the Euclidean norm. The reported order uses the three finest levels,

$$p_Q = \log_2 \frac{\|Q_{10} - Q_5\|}{\|Q_5 - Q_{2.5}\|}.$$

Only an exactly or roundoff-unchanged three-level sequence satisfies the flat-sequence alternative and is reported as such rather than as an invented infinite order. A flat finest pair following a nonzero coarse change fails this order component as possible stagnation or quantization.

4 Verification metrics and acceptance criteria

All four input-matched simulation conditions and their exact-stop TOA, total-DFC, and total-radiation rasters are mandatory. Each concrete namelist and prepared raster geometry must agree with the case specification and simulation specification. Each feedback log must contain every expected start time from 0 through $t_{stop} - \Delta t$, with maximum absolute error 0.051 s to accommodate the log's 0.1 s print precision. Finest-pair relative changes must not exceed 5% for landmark TOA and community ROS or 10% for total heat. A resolvable nonzero sequence must have observed order at least 0.5; a roundoff-flat three-level sequence is the explicit alternative. Every TOA fit must have $R^2 \geq 0.95$.

These are preregistered numerical-resolution criteria, not tolerances quoted by the references. Five percent makes a propagation observable materially smaller than a one-cell coarse-grid displacement; the 10-percent heat limit is looser because source/receiver overlap changes discontinuously with rasterization. The positive order floor of 0.5 detects contraction without asserting a formal order for this coupled, clipped model, and $R^2 = 0.95$ prevents a visibly nonlinear or fragmentary TOA profile from being summarized by a single ROS. The limits are not adjusted to the result.

5 Simulation configuration and predicted outcome

The physical community is 160 m by 40 m; every interior cell is fire behavior fuel model (FBFM) 91 and uses a constant 20 m linear characteristic building dimension, 10 m separation, and building fuel model 14. A 20 m-wide upwind strip is initially burning. The wind is 15 mph from the west, the design fire reaches 800 kW m^{-2} at 30 s, and the run ends at 300 s. Firebrands, crown fire, stochastic wind, and suppression are disabled. A point ignition scheduled after the stop time prevents the generic early-stall shortcut without affecting the solution.

Quantity	Values	Units	Control intent
Grid spacing	20, 10, 5, and 2.5 m	m	Ratio-two refinement
Timestep / stop	0.25 / 300	s	Isolates spatial refinement

Physical source	20 by 40	m	Equal initial burning area
Wind / direction	15 / 270	mph / degree from	Fixed ellipse forcing
WU-E band	Entire physical domain	cells	Equal heat-interaction support
Terminal evidence	TOA, total DFC, total radiation	s / kJ	Propagation and energy convergence

Figure 1 is generated directly from the prepared 5 m GeoTIFFs. It appears before any result figure so the fuel field and initial level-set support can be audited independently of simulation output.

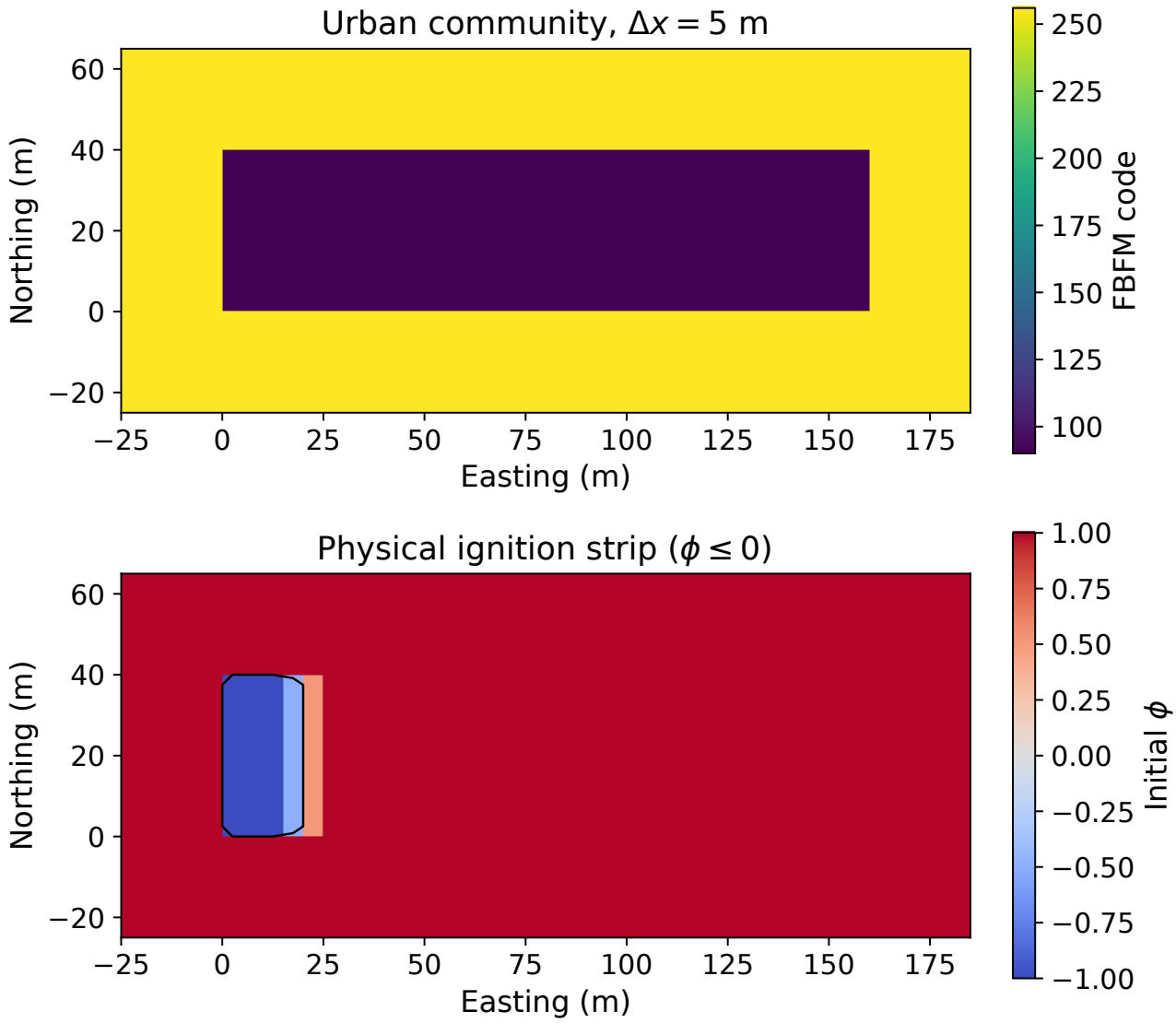


Figure 1: Mechanically rendered fuel-model classification and level-set field; this is the representative 5-m prepared grid, not a simulation result.

6 Actual simulation results

The analysis procedure state is **INCOMPLETE**; the scientific decision is **NOT EVALUABLE**, with 0 of 4 required simulation conditions. Reason: ELMFIRE execution was attempted, but one or more current-input identity record simulation conditions lack valid required evidence.. The executed binary SHA-256 is −−. For each completed simulation, the selected fields retain their actual output times and matching model, input, and simulation-record identities. All results must belong to those completed simulations.

7 Calculated metrics and verification decision

Current-input identity record output completeness — NOT EVALUABLE	
Criterion	4
Observed	*Units: simulation conditions.*
	0

The overall scientific decision is **NOT EVALUABLE**. Missing, stale, nonterminal, or malformed output is NOT EVALUABLE; NOT RUN is retained only as the analysis procedure state and can never produce a pass.

8 Technical interpretation

All required grids have complete terminal rasters and the prescribed solver step sequences. The failed finest-pair and observed-order rows therefore represent sensitivity of the calculated solution, rather than absent files or an unintended temporal refinement. The fitted TOA profiles remain suitable for the reported community ROS, but that fitted ROS does not meet the registered grid-convergence criterion.

The source contains grid-dependent heat normalization, overlap/support rules, and an energy-dependent replacement of material FTP. These are plausible contributors to the observed sensitivity. The finite sequence identifies a failed convergence test; it does not uniquely assign causality or prove asymptotic divergence. The existing limits remain unchanged.

9 Scope and limitations

This is a numerical verification case for an idealized, fully urban, uniform community. It does not validate building-to-building fire behavior against experiments, represent explicit street gaps or variable structure footprints, exercise firebrands, or cover wildland–urban interfaces. CASE46 and CASE47 address transition pathways; CASE43–CASE45 isolate the component responses used here.

References

[1] D. M. J. Purnomo et al., “Integrating an urban fire model into an operational wildland fire model to simulate one dimensional wildland–urban interface fires: a parametric study,” *International Journal of Wildland Fire*, 33 (2024), WF24102. doi:10.1071/WF24102.